

Respiratory Physiology: Shunt and V/Q

Shunt physiology causes arterial hypoxemia that responds incompletely to increased inspired oxygen because perfused alveoli are not ventilated. Low V/Q regions improve more with oxygen than true shunt. In the operating room, atelectasis, mainstem intubation, aspiration, and pneumonia are common causes of hypoxemia after induction.

Succinylcholine contraindications include major burns after 24 to 48 hours, denervation injury, neuromuscular disease, chronic immobilization, and personal or family history of malignant hyperthermia. Hyperkalemia is the board trap.

Local anesthetic systemic toxicity treatment includes airway support, seizure control, avoiding large vasopressin doses, and lipid emulsion therapy. Malignant hyperthermia management includes stopping triggering agents, calling for help, hyperventilating with 100 percent oxygen, giving dantrolene, active cooling, and treating hyperkalemia and acidosis.

MAC decreases with age, pregnancy, hypothermia, acute alcohol intoxication, opioids, benzodiazepines, and alpha-2 agonists. MAC increases with hyperthermia, chronic alcohol use, and stimulants.